

INVESTIGATION REPORT

Issue 1004: Interrupt during the first fetch after wake-up

openhwgroup/cv32e40p

Document	RTL-TRIAGE-CV32E40P-1004
Disposition	Reachable, directed test supplied
Record date	2026-09-11
Classification	Internal working document. Not a verification sign-off.

1 Status

Issue	openhwgroup/cv32e40p#1004
Issue type	Coverage hole referred for disposition
Affects	cv32e40p
Configuration	COREV_PULP=1, FPU=0 (RTL master 6033d2b)
Disposition	Reachable, directed test supplied
Severity	Not assessed
Agreement	The human record stated no proof or test. This disposition agrees with everything the record did say and adds evidence.
Evidence	6 cited, 5 present, 1 not retained

Disposition meaning. The code is reachable, and a directed test that reaches it is supplied.

Severity is not recorded for this run. The harness does not infer a severity from the disposition, so the field is reported as not assessed rather than filled in.

2 Description

cv32e40p_controller.sv:399 (FIRST_FETCH: if (irq_req_ctrl_i && ~(debug_req_pending || debug_mode_q))) has two uncovered conditions and they get different dispositions. (a) irq_req_ctrl_i && debug_mode_q in FIRST_FETCH is unreachable: abc pdr proves it unbounded in 85 s. FIRST_FETCH is entered only from BOOT_SET (debug_mode_q is 0 after reset and BOOT_SET with a pending debug request goes to DBG_TAKEN_IF instead) or from SLEEP, and the core never sleeps in debug mode (wfi is a nop when debug_wfi_no_sleep_o, which includes debug_mode_q). (b) irq_req_ctrl_i && debug_req_pending in FIRST_FETCH is reachable: the cover engine reaches it at step 16 through the sleep path (mstatus.MIE set, wfi -> WAIT_SLEEP -> SLEEP, wake on irq_i, then debug_req_i rises in the FIRST_FETCH cycle itself; a debug request one cycle earlier would have sent SLEEP to DBG_TAKEN_IF). The unconstrained pdr attempt on (b) timed out at 30 min without finding the trace (kept), the cover found it. A directed test (first_fetch_dt: timer interrupt wake from wfi, testbench debug_req pulse with a swept start delay, 32 iterations) reproduces the exact cycle on the core-v-verif Verilator testbench: 1 hit of FIRST_FETCH && irq_req_ctrl_i && debug_req_pending at cycle 13562, 0 hits of irq && debug_mode_q, every iteration takes both the debug entry and the interrupt afterwards, EXIT SUCCESS. Making that test run exposed two Verilator core-tb limitations (irq_i tied to zero in cv32e40p_tb_wrapper; instruction address truncated to 22 bits before mm_ram so debug code at 0x1A110800 is fetched from 0x110800); both are handled in setup.sh and flows/sim.sh without touching RTL, and interrupt_test from the regression benefits. Disposition: waive condition (a) as proven dead; do not waive (b), cover it with the directed test (a UVM equivalent needs a debug_req one cycle after the wfi wake).

3 Conditions

Not recorded as separate fields for this run. The conditions and environment assumptions that bound this result are stated inside the Description, and the configuration is given in Status.

4 Implications

Not recorded as a separate field for this run. What the disposition means for a user of the core follows from the Disposition meaning in Status and the bounds in Limitations and scope.

5 Action

Waiver text, to be applied to the coverage exclusion

cv32e40p_controller.sv:399 condition irq_req_ctrl_i && debug_mode_q in FIRST_FETCH: formally unreachable (abc pdr unbounded proof, runs/v1-cv32e40p-1004/formal/p_dbgmode); FIRST_FETCH is only entered from BOOT_SET or SLEEP and the core never sleeps in debug mode. Exclude from condition coverage. The other condition, irq_req_ctrl_i && debug_req_pending in FIRST_FETCH, is reachable (wfi wake on irq with debug_req arriving in the following cycle) and must NOT be waived: covered by directed test runs/v1-cv32e40p-1004/tests/first_fetch_dt (1 hit at cycle 13562 on the Verilator core tb).

Directed test

```
tests/first_fetch_dt/{first_fetch_dt.c, dbg.S}
```

Checker used

```
checker_1004.sv
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6 Evidence

Ref	Type	Path, relative to the run directory	File	Result recorded
E1	proof	formal/p_dbgmode/sby.log	present	abc pdr PASS unbounded (85.5 s): FIRST_FETCH && irq_req_ctrl_i -> !debug_mode_q
E2	proof	formal/cover/sby.log	present	cover: FIRST_FETCH&&irq at step 16, FIRST_FETCH&&debug_req_pending at step 5, FIRST_FETCH&&irq&&debug_req_pending (hole b) at step 16; irq&&debug_mode_q unreached to depth 40 (consistent with proof)
E3	counterexample	formal/cover/job/engine_0/trace1.vcd	not kept	hole (b) witness: MIE set at step 7, wfi, WAIT_SLEEP/SLEEP at 13-14, irq wake, FIRST_FETCH at 15 with debug_req_i=1 (cover_hole_line399_irq_and_debug_req.png)
E4	proof	formal/p_dbgreq/sby.log	present	negative result kept: unconstrained abc pdr on (b) TIMEOUT 30 min (property is false; cover produced the witness)
E5	simulation	sim/first_fetch_dt/sim.log	present	Verilator directed test first_fetch_dt: EXIT SUCCESS, 32 iterations, irq=32 dbg=32; vcd_hits: 1 hit of FIRST_FETCH&&irq&&debug_req_pending at t=135620 (cycle 13562), 0 hits of irq&&debug_mode_q
E6	waveform	sim_first_fetch_hit.png	present	simulation waveform of the hit cycle: SLEEP -> FIRST_FETCH with irq_req_ctrl_i and debug_req_pending both 1, interrupt held back, debug entry next

Paths are relative to runs/v1-cv32e40p-1004. The File column is a check made when this document was generated. It reports whether the cited file exists, not whether its contents were re-validated. 'not kept' means a repository rule deliberately excludes that artefact, usually a waveform dump; rerun the check to regenerate it. 'MISSING' means the file is absent and was expected to be there.

What each class of artefact establishes

- proof. Proof or engine run. Scope depends on the mode: prove is unbounded under the stated model, bmc is bounded to its depth, cover establishes reachability only.
- counterexample. A concrete input sequence produced by the engine. Shows the stated condition is reachable; it is not itself a proof.

Artefact(s) E3 are not retained under repository policy. The result recorded against them stands on the engine log cited

alongside; the artefact itself must be regenerated by rerunning the check before it can be inspected.

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Limitations and scope

- A directed test establishes that the target is reachable. It does not establish coverage of the surrounding logic or the absence of other holes.
- This is an agent investigation record. It requires engineering review before it is used as a disposition. No report generator can supply missing evidence.
- Results apply to the configuration and core revision named in Status. A different configuration needs a separate run.
- Measured tool time is the time the tools ran. It is not an engineering effort figure and not a turnaround commitment.
- Public demonstration runs on open cores are not client engagements and not production verification results.

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Provenance and measurement

Run directory	runs/v1-cv32e40p-1004
Title source	Title and configuration read back from reports/issues/cv32e40p_issue_1004.pdf, the previously published report for this issue.
Inputs shown	title, body(links stripped), labels
Human record	Dolphin Design 2024 (issue body): 'was waived in v1 and has been waived in v2 as well', no proof or test
Agent confidence	0.85. This is the investigating agent's own estimate. It is not a measurement and not a reliability figure.
Iterations	5
Run by	Claude directly (no headless agent)

Measured tool and step time

Step	Measured
read issue + FSM analysis (two conditions)	7.0 min (420 s)
asserts + 4 covers, pdr + cover	4.6 min (275 s)
read witness: sleep/wake/debug_req recipe	5.0 min (300 s)
directed test v1-v4 (stdout buffering, .debugger load address, irq_i tied off, compressed vector table)	90.0 min (5400 s)
directed test hit + waveform	2.0 min (120 s)
report	7.0 min (420 s)
Total	115.6 min (6935 s)

Times are wall-clock for the step named. Some steps may have run concurrently, so they do not sum to an isolated performance figure, and they are not an engineering effort measure.

Generated by reporting/scripts/issue_document.py from runs/v1-cv32e40p-1004/verdict.json. The generator typesets recorded data and checks that cited files exist. It does not run, rerun or validate any check, and it cannot supply missing evidence.